

John Aledare

Machine Learning Engineer — Computer Vision & Deep Learning

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EDUCATION

University of Ilorin

Ilorin, Nigeria

Bachelor of Engineering (B.Eng.) in Computer Engineering

2021 – 2026

- Relevant Coursework: Data Structures & Algorithms, Computer Architecture, Operating Systems, Intelligent Systems, Signal Processing, Software Engineering.

CERTIFICATIONS

OCI Generative AI Professional | Oracle

2024

Oracle AI Foundations Associate | Oracle

2024

Machine Learning Specialization | Stanford University & DeepLearning.AI

2024

TECHNICAL SKILLS

Languages: Python, TypeScript

Deep Learning Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, OpenCV

Model Optimization & Deployment: ONNX, ONNX Runtime, Model Quantization (INT8 / FP16), TensorFlow Lite, Transfer Learning, Edge Deployment, Docker, FastAPI, Streamlit

ML Domains: Computer Vision, Deep Learning, Segmentation, Depth Estimation, Perception, NLP, MLOps & Model Serving, Reinforcement Learning

Data & Analytics: NumPy, Pandas, Exploratory Data Analysis, Hypothesis Testing, Metric Trees, Power BI (DAX, Dashboards, KPI Reporting)

Infrastructure & DevOps: Docker, Linux, Git, Pydantic v2, REST APIs

EXPERIENCE

Queryfier LLC

Remote

Machine Learning Engineer

Jan 2026 – Present

- Architect, train, and deploy production-grade machine learning models for NLP and Computer Vision tasks, owning the full development lifecycle from prototyping to production serving.
- Construct end-to-end data preprocessing and continuous training pipelines using Scikit-learn, TensorFlow, and PyTorch to support large-scale model development and evaluation.
- Build low-latency asynchronous API inference microservices with FastAPI and interactive validation interfaces using Streamlit, enabling real-time model testing and stakeholder validation.
- Enforce schema validation (Pydantic v2), containerize services with Docker, and implement real-time model monitoring to standardize ML microservice deployment and runtime data integrity.
- Deployed production model serving architecture handling asynchronous multimodal payloads, supporting both NLP and Computer Vision inference workloads.

Centre for Applied Machine Learning and Data Science (CAMLDS)

On-site

Machine Learning Engineer Intern & Python Tutor

Mar 2025 – Dec 2025

- Developed and evaluated supervised and unsupervised machine learning pipelines for text classification and computer vision tasks, applying rigorous testing and metric analysis across datasets.
- Built and deployed web demo prototypes using Streamlit and FastAPI for model validation, enabling interactive evaluation of trained models.
- Trained, validated, and packaged baseline ML models for research initiatives, developing robust data pipelines using NumPy, Pandas, and Scikit-learn.
- Mentored junior interns and led hands-on tutoring sessions in Python, NumPy, Pandas, Scikit-learn, and neural networks, communicating complex ML concepts to interdisciplinary audiences.

Technical Mentorship & Community Volunteering

Remote

Python & Machine Learning Mentor

2023 – Present

- Organize and lead hands-on workshops on Python, data structures, and foundational machine learning, providing code reviews and debugging guidance for aspiring AI/ML engineers.
- Author educational articles on Medium explaining complex AI and ML concepts in plain English, strengthening technical communication skills across interdisciplinary topics.

PROJECTS

Intelligent System for Lawn Mowers (Autonomous Intelligent Lawn Mower) | *Python, YOLO, TensorFlow Lite, INT8*

- Designed a computer-vision perception pipeline spanning training dataset preparation, YOLO training, model evaluation, conversion to TensorFlow Lite, INT8 quantization, and real-time inference on Raspberry Pi.
- Engineered the pipeline specifically for edge deployment constraints, ensuring the model was not limited to desktop-only inference but optimized for real-time operation on constrained hardware.

Brain Tumor MRI Classifier | *Python, TensorFlow, EfficientNetB0, Transfer Learning, INT8 Quantization, Grad-CAM, Streamlit*

- Trained deep learning medical vision classifiers using EfficientNetB0 transfer learning with Grad-CAM visual heatmaps for clinical interpretability, successfully classifying four diagnostic classes (Glioma, Meningioma, Pituitary, Normal).
- Applied model quantization for edge deployment on mobile applications and interactive Streamlit web apps, reducing model parameter size for low-latency inference.
- Evaluated model performance through confusion matrix analysis and deployed a Streamlit application for interactive validation.

Flappy Bird Reinforcement Learning & Benchmark | *Python, PyTorch, PPO, NEAT Neuroevolution, Gymnasium, Tabular Q-Learning*

- Implemented and benchmarked multiple reinforcement learning approaches including Proximal Policy Optimization (PPO), NEAT neuroevolution, and Tabular Q-Learning using Gymnasium and PyTorch.
- Built a local benchmark suite to systematically compare RL algorithm performance across training runs.

Nigerian Pidgin Next-Word Predictor | *Python, PyTorch, LSTM, Trigram Statistical Model, Real-time Inference*

- Trained a deep LSTM recurrent neural network alongside a Trigram statistical model to predict next words in Nigerian Pidgin English, balancing deep neural network accuracy against statistical n-gram speed.
- Implemented keystroke debouncing for real-time inference over an interactive UI, addressing the lack of robust language models for low-resource languages.

AI Resume Optimizer & Career Architect | *Python, TypeScript, ONNX Quantized NER Model, FastAPI, Next.js, Docker, Vercel*

- Integrated an ONNX quantized NER model with FastAPI and Next.js to build a full-stack ML pipeline from data ingestion through model inference and deployment.
- Containerized and deployed the service using Docker and Vercel, demonstrating end-to-end ML pipeline deployment across cloud infrastructure.

PUBLICATIONS & TECHNICAL WRITING

Artificial Intelligence in Plain English (Medium Publication)

- **How Do Neural Networks Actually "Learn"?:** Intuitive mathematical guide explaining weight optimization, cost surfaces, gradient descent algorithms (SGD, Adam, RMSProp), computational graphs, and the calculus behind backpropagation with clear step-by-step visualizations.
- **ChatGPT Will Agree That 2+2=5 If You Ask Nicely Enough:** In-depth analysis of sycophancy in LLMs, Reinforcement Learning from Human Feedback (RLHF) reward hacking, vulnerabilities in conversational fact-checking, and how Constitutional AI and automated red-teaming help mitigate agreeable bias.